

EN3021 Individual Project - Progress as of OCT03

210609M Silva M.K.Y.U.N.

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Current Status: 30% complete

Completed Parts

Following parts of the project have been completed as of October 03:

- The paper design of adder and subtractor
- CLA and Shifter Logic (tested with a testbench)
- Started the Verilog design for adder and subtractor parts

Paper Design

Pipelines

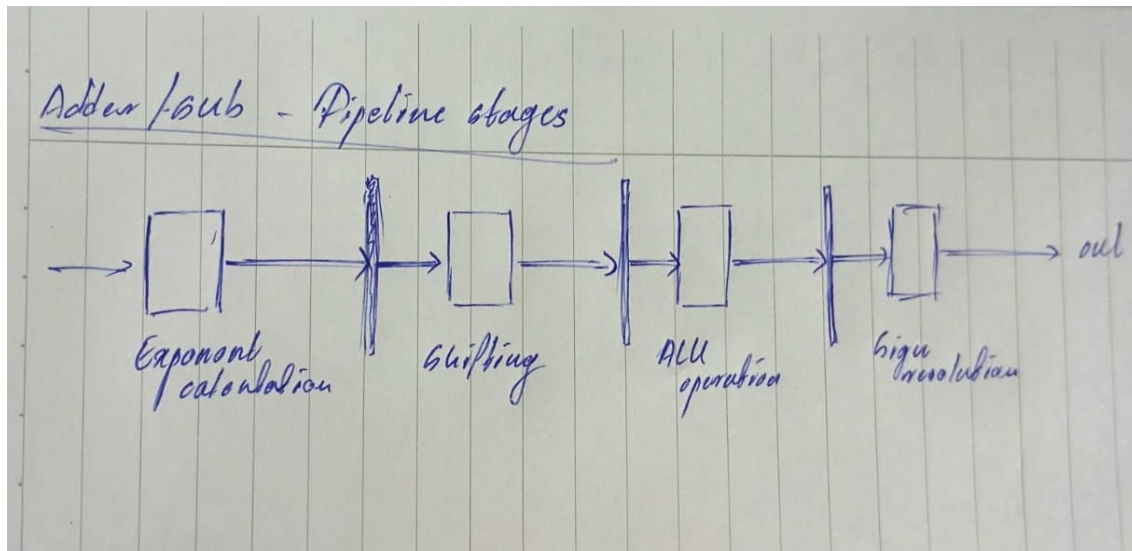


Figure 1: Pipelines for the adder/sub

Here there are 4 stages in the pipeline for adder and subtractor:

- Stage1: Exponent difference
- Stage2: Shifting
- Stage3: ALU Operation
- Stage4: Sign resolution

Basic Block Diagram

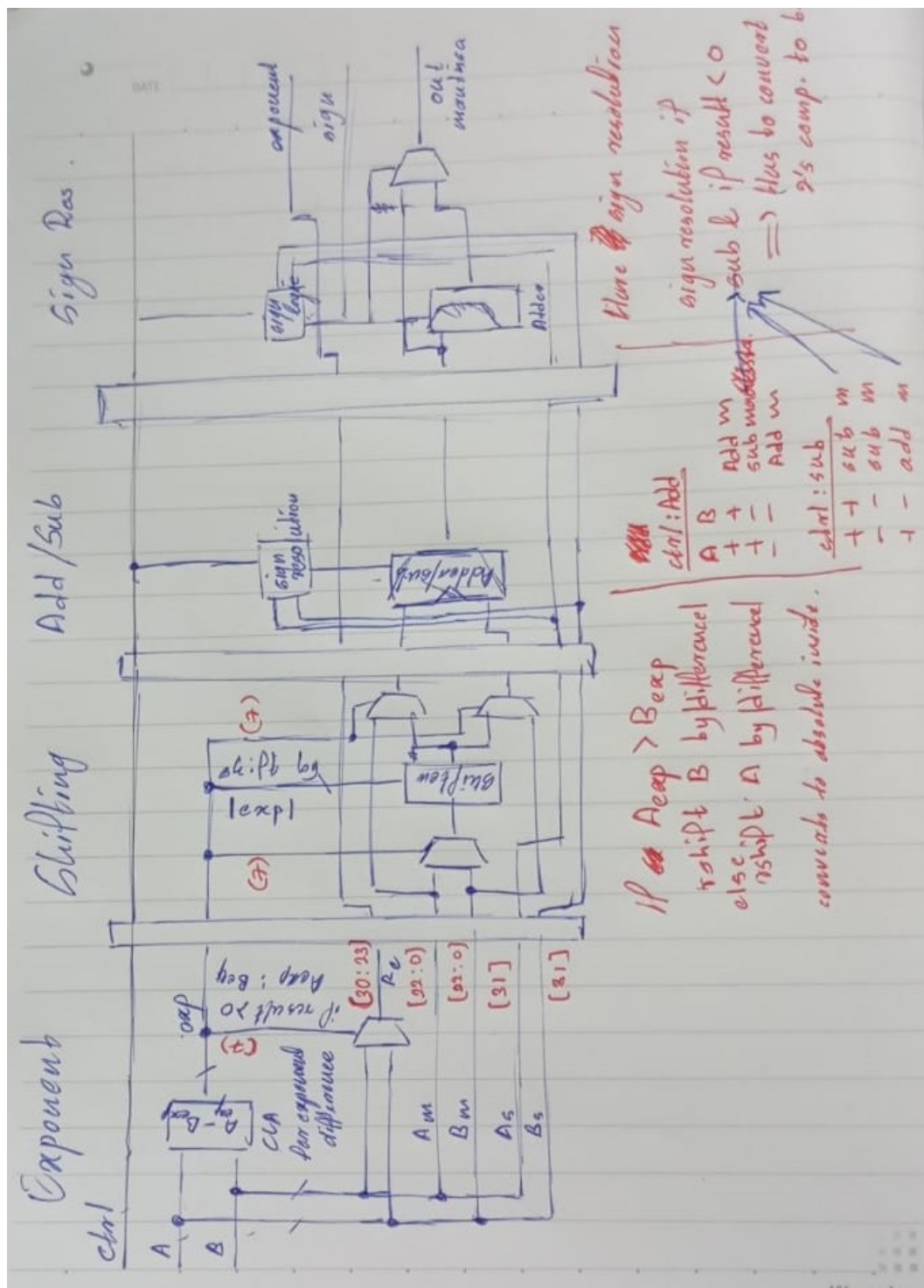


Figure 2: Paper design for add/sub part

CLA and Shifter

CLA

```

module cla(
    input  [3:0] A, B,
    input  CIN,
    output [3:0] COUT,
    output [3:0] S
);
    // For carry calculation

    wire [3:0] P, G;
    assign P = A ^ B;
    assign G = A & B;

    wire [3:0] CARRY;

    assign CARRY[0] = G[0] | (P[0] & CIN); // CARRY[0]
    assign CARRY[1] = G[1] | (P[1] & G[0]) | (P[1] & P[0] & CIN); // CARRY[1]
    assign CARRY[2] = G[2] | (P[2] & G[1]) | (P[2] & P[1] & G[0]) | (P[2] & P[1] & P[0] & CIN); // CARRY[2]
    assign CARRY[3] = G[3] | (P[3] & G[2]) | (P[3] & P[2] & G[1]) | (P[3] & P[2] & P[1] & G[0]) | (P[3] & P[2] & P[1] & P[0] & CIN); // CARRY[3]

    assign COUT = CARRY[3];

    // Adder
    assign S = P ^ {CARRY[2:0], CIN};
endmodule

```

Figure 3: Verilog Design for CLA

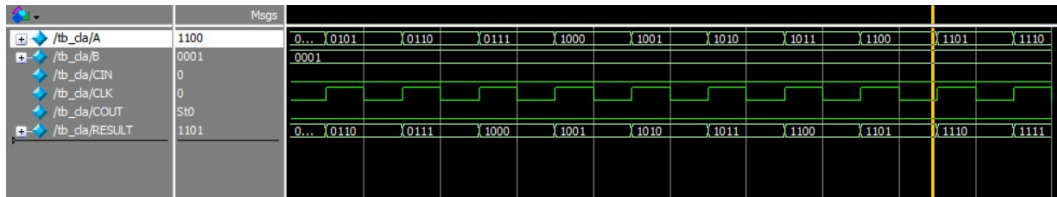


Figure 4: Testbench simulation

Shifter

Here the shifter is designed such that we could input the difference between the exponents of A and B ($A[30:23] - B[30:23]$) as it is and the shifter would shift the input to right by the absolute value of that input.

```

1 module shifter(
2     input  [23:0] IN,
3     input  [7:0] BY,
4     output reg [23:0] OUT
5 );
6
7     wire COUT_INT;
8     wire [7:0] BY_COMPL;
9
10    // Calculating the absolute
11
12    cla cla_int_sh0(
13        .A(~BY[3:0]),
14        .B(4'b1),
15        .CIN(1'b1),
16        .COUT(COUT_INT),
17        .S(BY_COMPL[3:0])
18    );
19
20    cla cla_int_sh1(
21        .A(~BY[7:4]),
22        .B(4'b0),
23        .CIN(COUT_INT),
24        .COUT(),
25        .S(BY_COMPL[7:4])
26    );
27
28    always@(IN or BY) begin
29        if (BY[7] == 1'b0) begin // When BY is > 0
30            OUT = IN >> BY;
31        end else begin // When BY is < 0
32            OUT = IN >> BY_COMPL;
33        end
34    end
35 endmodule

```

Figure 5: Verilog Design for Shifter

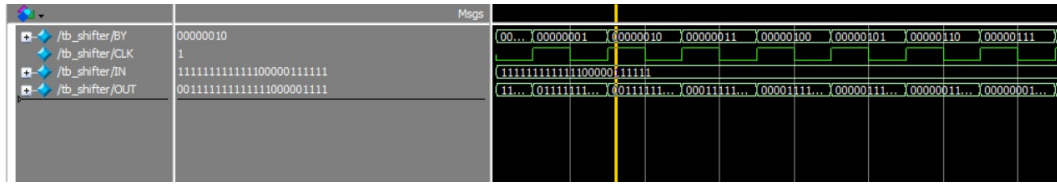


Figure 6: Testbench simulation